Case Report

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Differential hypoxemia during peripheral cardiopulmonary bypass

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Abstract

Introduction: Minimally invasive cardiac surgery (MICS) for mitral valve repair often uses cardiopulmonary bypass (CPB) through peripheral femoro-femoral cannulation. A rare complication of differential hypoxemia can cause upper body hypoxia.

Case Report: A 38-year-old man with Barlow Syndrome underwent MICS mitral valve repair with peripheral CPB. During the procedure, his right upper limb SpO2 dropped to 65% due to dislodgement of the venous cannula from the superior vena cava (SVC), causing inadequate right heart venous drainage, leading to differential hypoxemia. Repositioning the venous cannula to the SVC restored SpO2 to 100%, allowing the surgery to proceed without complications.

Discussion: Differential hypoxemia happens when deoxygenated blood from the left ventricle mixes with oxygenated CPB blood. A literature review found only one previous case of this complication on peripheral CPB for MICS. Unlike our case, Kanda et al. monitored cerebral hypoxia, underscoring the importance of cerebral oxygenation monitoring in future MICS procedures due to higher stroke risk compared to conventional sternotomy.

Conclusion: We recommend cerebral oxygenation monitoring in future MICS to prevent neurological complications, and highlight the need for effective team communication in the cardiac operating theatre.

Keywords

minimally invasive cardiac surgery (MICS), differential hypoxemia, mitral valve repair, peripheral cardiopulmonary bypass

Introduction

Minimally invasive cardiac surgery (MICS) for mitral valve repair has become a common treatment for symptomatic mitral regurgitation, typically using cardiopulmonary bypass (CPB) through peripheral femoro-femoral cannulation.[1] This method perfuses the body with oxygenated blood in a retrograde fashion, allowing lung deflation for a prolonged period of time for surgical purposes. However, incomplete right heart drainage on CPB whilst the patient's cardiac output is still present, can lead to deoxygenated blood entering the arterial circulation, creating a watershed zone where oxygenated and deoxygenated blood mix in the aorta.[2] This can cause hypoxia in areas such as the coronary and carotid arteries, leading to heart and brain hypoxia, and upper limb hypoxaemia.[3]

This case study reports a young man experiencing right upper limb hypoxaemia during MICS for mitral valve repair due to such differential hypoxemia.

Case study

A 38 year old gentleman with Barlow Syndrome and a New York Heart Association (NYHA) score of III, underwent elective MICS mitral valve repair with peripheral femoro-femoral CPB. Written consent was obtained. The patient had presented with a New York Heart Association (NYHA) classification score of III.

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Preoperative transesophageal echocardiography (TEE) showed severe mitral valve prolapse, annular dilation, and severe insufficiency. The patient, otherwise healthy, weighed 79 kg and was 171 cm tall.

Anaesthesia was induced using intravenous 10 mg of midazolam, 30 mcg of sufentanil, 30 mg of propofol and 80 mg of rocuronium. Anaesthesia was maintained with an infusion of propofol and sufentanil, and sevoflurane/oxygen/air gas mixture. Bispectral index of the patient was monitored throughout the procedure. Peripheral oxygen saturation (SpO2) was continuously monitored using pulse oximetry on the right hand. A right brachial arterial line was placed prior to induction of anaesthesia. Peripheral CPB was established with cannulation of the femoral artery and femoral vein. The femoral venous cannula used was a Maquet Venous HLS cannula 25 Fr (REF PVL2555) without coating. The tip of the venous cannula was placed in the superior vena cava (SVC) and confirmed on TEE. Mechanical ventilation was stopped after target flow on the CPB was reached at 4610 L/min at a cardiac index of 2.4. VAVD was used at a maximum of - 60 mmHg suction pressure. Shortly after mechanical ventilation was stopped, prior to cardioplegia administration, the patient's SpO2 dropped to 65%. The reservoir volume of the CPB was noted to be decreased, reflecting a decrease in venous drainage, whilst CPB flow was stable and a colour change of the blood passing through the oxygenator was observed, indicating adequate oxygenator function. On TEE, the venous cannula was noted to have been dislodged with its tip in the inferior vena cava (IVC). Simultaneously, an arterial blood gas analysis from the right brachial artery and from the CPB machine were taken during this desaturation. The right brachial artery blood gas showed a partial pressure of oxygen (pO2) of 38 mmHg and an oxygen saturation (SaO2) of 64.8%. The CPB arterial blood gas showed a pO2 of 318 mmHg and an SaO2 of 99.3% (Table 1). As soon as the dislodgement was noted,the tip of the venous cannula was repositioned into the superior vena cava (SVC) andconfirmed on TEE. The SpO2 of the right upper limb recovered from 65% to 100% shortly after repositioning of the venous cannula. No changes on the electrocardiogram were noted, including ST-segment alterations, during the period of desaturation. Cardioplegia was given and the surgical procedure was performed without any further complications. The patient was then transferred to the cardiac intensive care unit and extubated the next day. He had no neurological sequelae and was well. The patient was discharged from hospital after 6 days. Verbal consent for this case report was obtained from the patient prior to discharge from hospital.

Table 1. Comparison of right brachial and cardiopulmonary bypass arterial blood gas analysis.

	Right brachial artery	CPB arterial sample
Time	16:24	16:23
pH	7.258	7.398
pO2	38.5 mmHg	318.1 mmHg
pCO2	51.9 mmHg	40.8 mmHg
HCO3-	22.7 mmHg	24.6 mmHg
BE	−4.7 mmol/L	−0.2 mmol/L
Hct	34%	33%
Hb	11.4 g/dL	11.2 g/dL
SaO2	64.80%	99.30%
Na+	131.6 mmol/L	131.4 mmol/L
K+	3.4 mmol/L	3.17 mmol/L
Ca2+	1.14 mmol/L	1.1 mmol/L
Cl-	98 mmol/L	99 mmol/L
Anion gap	14.3 mmol/L	11 mmol/L
Glucose	125 mg/dL	130 mg/dL
Lactate	0.81 mmol/L	0.93 mmol/L

Abbreviations: CPB, cardiopulmonary bypass; p, partial pressure; BE, base excess; Hct, haematocrit; Hb, haemoglobin; SaO2, oxygen saturation.

Discussion

An extensive literature review revealed only one previous case of differential hypoxemia on peripheral CPB for MICS.[4] In this previous case report, differential hypoxemia occurred due to dislodgement of the venous cannula leading to inadequate venous drainage, similar to our case study. However, Kanda et al. also monitored the patient for cerebral hypoxia using laser speckle flowgraphy (LSFG) and near infrared spectroscopy (NIRS). Therefore, the lack of cerebral hypoxia monitoring was a limitation in our case report and we recommend it to be used in future MICS on peripheral CPB. This recommendation is supported by a significantly higher risk of strokes in MICS as compared to conventional sternotomy.[5] One might also consider preferential right upper limb oxygen saturation monitoring for early detection of differential hypoxemia. If the coronary arteries are also perfused with deoxygenated blood, ischaemia and cardiac dysfunction may occur.

In our case study, upper body hypoxaemia resulted from differential hypoxemia due to inadequate venous drainage from a dislodged venous catheter, a preserved cardiac output, and suspended mechanical ventilation. Deoxygenated blood from the left ventricle mixed with oxygenated blood from the CPB machine. Once the venous cannula was repositioned to the SVC, the patient's right arm SpO2 returned to 100%. This case highlights the need for proper venous drainage from the right heart in MICS for mitral valve repair before aortic clamping. TEE routinely confirms SVC cannula placement, and additional drainage cannulas in the internal jugular can be used if femoral drainage is inadequate.[2]

In conclusion, this case highlights a rare but significant complication of differential hypoxemia during MICS for mitral valve repair on peripheral CPB. Inadequate venous drainage, preserved cardiac output and suspended mechanical ventilation led to upper body hypoxaemia. We recommend the use of cerebral oxygenation monitoring for cerebral hypoxia in future MICS procedures to prevent neurological complications and emphasise the need for effective communication among the cardiac operating theatre team.

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